SHOWSHELF LAB · FOR THE AFTERNOON TEACHER

Teacher guide: Day 1

Students work through the steps in the Lab on their own. Each step has an explanation, a task, two hints and an answer button, so most of the time they can get unstuck alone. This guide gives you every task, a correct answer and the mistakes students usually make, so you can help quickly.

How to help: first ask them to read the "Not yet" message and the red Console line (if any), then open a hint. Only point at the answer after that. In the Lab, **Help › Teacher view** shows the answer and these notes on every step.

Day 1: Your first web page HTML

Today you build a web page about one TV show, **Friends**, using **HTML**: the language that tells the browser what each piece of a page *is*.

You start with an empty box and add one thing at a time: a heading, a paragraph, a picture, lists, a link and a table. The top bar and the footer are already there, so you can focus on the show itself.

Take your time with each step. There's no rush, and every step saves automatically.

Step 1. Your first HTML tag ~10 min

Teaches: An element is an opening tag, content and a closing tag. `<h1>` is the main heading.

TASK

In the dark code box, write a main heading that says **Friends**.

Watch the preview on the right while you type. Then press **Check**.

COMMON MISTAKES

- Forgetting the slash in the closing tag: `<h1>Friends<h1>`.
- A space inside the brackets: `< h1>`.
- Writing in the preview instead of the dark code box.

NOTE

The goal is only the idea of opening tag, content, closing tag. Let them try breaking it: delete the closing tag and see what happens (the browser guesses).

A CORRECT ANSWER

```
<h1>Friends</h1>
```

Step 2. Add a paragraph ~10 min

Teaches: `<p>` makes a paragraph. The browser ignores extra spaces and new lines in your code.

TASK

Under your heading, add a paragraph with this text (copy it):

Six friends in their twenties and thirties share apartments, jobs, break-ups and a lot of coffee in New York City.

COMMON MISTAKES

- Putting the paragraph inside the heading: `<h1>Friends <p>...</p></h1>`.
- Forgetting `</p>` (the browser often guesses, but the check asks for it).

A CORRECT ANSWER

```
<h1>Friends</h1>

<p>
  Six friends in their twenties and thirties share apartments, jobs,
  break-ups and a lot of coffee in New York City.
</p>
```

Step 3. Bold and italic ~10 min

Teaches: Elements can be nested. `<strong>` is important (bold), `<em>` is emphasis (italic).

TASK

In your paragraph:

1. make **New York City** bold with `<strong>`
2. make **a lot of coffee** italic with `<em>`

COMMON MISTAKES

- Closing in the wrong order: `<p>...<strong>...</p></strong>`.
- Using `<b>` and `<i>`. They also work, but `<strong>` and `<em>` say *why* the words are special.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
<p>
  Six friends in their twenties and thirties share apartments, jobs,
  break-ups and <em>a lot of coffee</em> in <strong>New York City</strong>.
</p>
```

Step 4. Attributes and a class ~15 min

Teaches: Attributes go in the opening tag as `name="value"`. A `class` is a label for CSS.

TASK

Between the heading and the paragraph, add a new paragraph with the class `meta` and this text:

NBC · 1994–2004 · about 22 minutes per episode

The dot `·` is hard to type on some keyboards. A normal dash `-` or a `|` is fine too.

COMMON MISTAKES

- No quotes: `class=meta` (works in browsers, but always use quotes).
- The attribute in the closing tag: `</p class="meta">`.

A CORRECT ANSWER

```
<h1>Friends</h1>
<p class="meta">NBC · 1994–2004 · about 22 minutes per episode</p>

<p>
<!-- ... unchanged ... -->
```

Step 5. Add the poster ~15 min

Teaches: `<img src="..." alt="...">` shows a picture. It has no closing tag, and `alt` describes the picture.

TASK

Under the heading, add the poster:

- `src` is `poster.svg`
- `alt` describes it, for example "Poster for Friends"
- also add `class="poster"`, for tomorrow's CSS

Then try this: change `src` to a name that doesn't exist, like `posterr.svg`. What do you see? Change it back before you press Check.

COMMON MISTAKES

- Writing `<img src="poster.svg"></img>`.
- Leaving out `alt`, or writing "image" as the alt text (describe what it shows instead).
- Typing `scr` instead of `src`.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
<p class="meta">NBC · 1994–2004 · about 22 minutes per episode</p>

<img class="poster" src="poster.svg" alt="Poster for Friends">

<p>
  Six friends in their twenties and thirties share apartments, jobs,
<!-- ... unchanged ... -->
```

Step 6. Subheadings and a list ~15 min

Teaches: `<h2>` for parts of the page. `<ul>` with `<li>` items makes a bulleted list.

TASK

Under your paragraph, add:

1. a subheading `<h2>` that says **Genres**
2. a bulleted list with two items: **Comedy** and **Romance**

COMMON MISTAKES

- `<li>` items without a `<ul>` around them.
- Two separate lists, one per item, instead of one list with two items.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
  break-ups and <em>a lot of coffee</em> in <strong>New York City</strong>.
</p>

<h2>Genres</h2>
<ul>
  <li>Comedy</li>
  <li>Romance</li>
</ul>
```

Step 7. A longer list: the cast ~10 min

Teaches: Long lists: one `<li>` per line, indented inside the `<ul>`.

TASK

Under the genres list, add an `<h2>` that says **Main cast** and a list with **at least three** of these actors:

- Jennifer Aniston as Rachel Green
- Courteney Cox as Monica Geller
- Lisa Kudrow as Phoebe Buffay
- Matt LeBlanc as Joey Tribbiani
- Matthew Perry as Chandler Bing
- David Schwimmer as Ross Geller

COMMON MISTAKES

- Putting all actors in one `<li>` separated by commas.
- Forgetting the new `<h2>`.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
  <li>Romance</li>
</ul>

<h2>Main cast</h2>
<ul>
  <li>Jennifer Aniston as Rachel Green</li>
  <li>Courteney Cox as Monica Geller</li>
  <li>Lisa Kudrow as Phoebe Buffay</li>
  <li>Matt LeBlanc as Joey Tribbiani</li>
  <li>Matthew Perry as Chandler Bing</li>
  <li>David Schwimmer as Ross Geller</li>
</ul>
```

Step 8. A numbered list ~10 min

Teaches: `<ol>` is a numbered list, for when the order matters.

TASK

Add an `<h2>` that says **My favourite characters** and an ordered list of your top three. Your opinion, your order.

COMMON MISTAKES

- Typing "1." inside each item (you get numbers twice).

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
  <li>David Schwimmer as Ross Geller</li>
</ul>

<h2>My favourite characters</h2>
<ol>
  <li>Phoebe</li>
  <li>Chandler</li>
  <li>Joey</li>
</ol>
```

Step 9. A link to another website ~15 min

Teaches: `<a href="https://...">text</a>` is a link to another page.

TASK

At the bottom, add a paragraph with a link to the show's page on TVmaze, a free website about TV shows:

`https://www.tvmaze.com/shows/431/friends`

Use any link text you like, for example "Read more on TVmaze". Click the link in the preview when you're done: it opens in a new tab.

COMMON MISTAKES

- Leaving out `https://`, so the link points to a missing file.
- Putting the address as the link text but forgetting `href`.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
  <li>Joey</li>
</ol>

<p>
  Want more? <a href="https://www.tvmaze.com/shows/431/friends">Read about it on TVmaze</a>.
</p>
```

Step 10. A table of seasons ~20 min

Teaches: Tables: `<table>`, rows `<tr>`, heading cells `<th>`, data cells `<td>`.

TASK

Add two more rows to the table:

- season **2**: **24** episodes, **1995**
- season **3**: **25** episodes, **1996**

Each new row is a `<tr>` with three `<td>` cells, just like the season 1 row.

COMMON MISTAKES

- Pasting the new row after `</table>` instead of before it.
- A row with only two `<td>`: the columns shift.
- Missing `</tr>` between rows.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
</ol>

<h2>Seasons</h2>
<table>
  <tr>
    <th>Season</th>
    <th>Episodes</th>
    <th>Year</th>
  </tr>
  <tr>
    <td>1</td>
    <td>24</td>
    <td>1994</td>
  </tr>
  <tr>
    <td>2</td>
    <td>24</td>
    <td>1995</td>
  </tr>
  <tr>
    <td>3</td>
    <td>25</td>
    <td>1996</td>
  </tr>
</table>

<p>
  Want more? <a href="https://www.tvmaze.com/shows/431/friends">Read about it on TVmaze</a>.
<!-- ... unchanged ... -->
```

Step 11. Group it with sections ~15 min

Teaches: Semantic elements (`header` , `main` , `article` , `section` , `footer`) describe the parts of a page.

TASK

Wrap **Genres** (the `<h2>` and its list) in a `<section>`. Then do the same for **Main cast**.

If you like, wrap the other two parts as well. Indent the lines inside each section by two spaces.

COMMON MISTAKES

- One big section around everything.
- `</section>` placed between the heading and its list.

A CORRECT ANSWER

```
<h1>Friends</h1>
<p class="meta">NBC · 1994–2004 · about 22 minutes per episode</p>

<img class="poster" src="poster.svg" alt="Poster for Friends">

<p>
  Six friends in their twenties and thirties share apartments, jobs,
  break-ups and <em>a lot of coffee</em> in <strong>New York City</strong>.
</p>

<section>
  <h2>Genres</h2>
  <ul>
    <li>Comedy</li>
    <li>Romance</li>
  </ul>
</section>

<section>
  <h2>Main cast</h2>
  <ul>
    <li>Jennifer Aniston as Rachel Green</li>
    <li>Courteney Cox as Monica Geller</li>
    <li>Lisa Kudrow as Phoebe Buffay</li>
    <li>Matt LeBlanc as Joey Tribbiani</li>
    <li>Matthew Perry as Chandler Bing</li>
    <li>David Schwimmer as Ross Geller</li>
  </ul>
</section>

<section>
  <h2>My favourite characters</h2>
  <ol>
    <li>Phoebe</li>
    <li>Chandler</li>
    <li>Joey</li>
  </ol>
</section>

<section>
  <h2>Seasons</h2>
  <table>
    <tr>
      <th>Season</th>
      <th>Episodes</th>
      <th>Year</th>
    </tr>
    <tr>
```

```
      <td>1</td>
      <td>24</td>
      <td>1994</td>
    </tr>
    <tr>
      <td>2</td>
      <td>24</td>
      <td>1995</td>
    </tr>
    <tr>
      <td>3</td>
      <td>25</td>
      <td>1996</td>
    </tr>
  </table>
</section>

<p>
  Want more? <a href="https://www.tvmaze.com/shows/431/friends">Read about it on TVmaze</a>.
</p>
```

Step 12. Notes for humans ~10 min

Teaches: `<!-- comments -->` are notes the browser ignores.

TASK

1. Add a comment above the seasons section that says what it is.
2. Try hiding the "My favourite characters" section with a comment, look at the preview, then remove the comment marks again so it comes back.

COMMON MISTAKES

- A comment that never closes (no `-->`): everything after it disappears.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
</section>

<!-- Seasons: one row per season -->
<section>
  <h2>Seasons</h2>
<!-- ... unchanged ... -->
```

Extra 1. A spoiler you have to click ~10 min

Teaches: `<details>` with `<summary>` makes a box that opens on click.

TASK

Add a `<details>` element with a `<summary>` like "Spoiler: how does the show end?" and a paragraph inside with your answer (or "Watch it and find out!").

Click it in the preview to open and close it.

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
  Want more? <a href="https://www.tvmaze.com/shows/431/friends">Read about it on TVmaze</a>.
</p>

<details>
  <summary>Spoiler: how does the show end?</summary>
  <p>Watch it and find out!</p>
</details>
```

Extra 2. A poster with a caption ~10 min

Teaches: `<figure>` + `<figcaption>`: a picture with a caption.

TASK

Wrap your poster `<img>` in a `<figure>` and add a `<figcaption>` under it, for example "The official poster".

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
<p class="meta">NBC · 1994–2004 · about 22 minutes per episode</p>

<figure>
  <img class="poster" src="poster.svg" alt="Poster for Friends">
  <figcaption>The official poster</figcaption>
</figure>

<p>
<!-- ... unchanged ... -->
```

Extra 3. Add a second show ~20 min

Teaches: You can build a page section from scratch: heading, text, list, link.

TASK

At the bottom, add:

1. an `<hr>`
2. an `<h2>` with the name of a show you like
3. a paragraph about it
4. a list of at least two things (genres, characters, anything)
5. optional: a link to its page on TVmaze (search for it on tvmaze.com)

A CORRECT ANSWER

```
<!-- ... unchanged ... -->
  Want more? <a href="https://www.tvmaze.com/shows/431/friends">Read about it on TVmaze</a>.
</p>

<hr>
<h2>The Office</h2>
<p>A camera crew follows the everyday chaos of a small paper company office.</p>
<ul>
  <li>Comedy</li>
  <li>Michael Scott is the boss</li>
</ul>
```